

Hands-On Lab 9:

Build ETL Pipeline Scenario Load customer data from:

CSV → Staging Table → DWH Table

Steps:

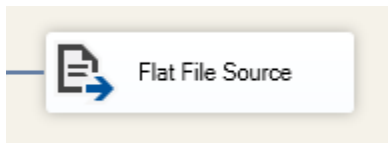
1. Load CSV
2. Clean data
3. Load to warehouse

Load CSV (Extract)

Add **Flat File Source**.

Steps

1. Open **Data Flow Task**.
2. Drag **Flat File Source**.
3. Double-click it.
4. Click **New** in Connection Manager.
5. Select the **CSV file path**.
6. Set **Column delimiter = comma (,)**.
7. Preview data.



Flat File Connection Manager Editor

Connection manager name: Flat File Connection Manager

Description:

General
Columns
Advanced
Preview

Select a file and specify the file properties and the file format.

File name: C:\Users\Administrator\Downloads\data_c Browse...

Locale: English (India) ☐ Unicode

Code page: 1252 (ANSI - Latin I)

Format: Delimited

Text qualifier: <none>

Header row delimiter: {CR}{LF}

Header rows to skip: 0

☒ Column names in the first data row

OK Cancel Help

Data View

Sample data (up to first 200 rows)

CustomerID	CustomerN...	City	Salary
1	Ravi	Chennai	5000
2	Arun	Bangalore	6000
3		Hyderabad	4500
4		Pune	0

2 Clean Data (Transform)

You already used three transformations:

A. Derived Column (Fix NULL values)

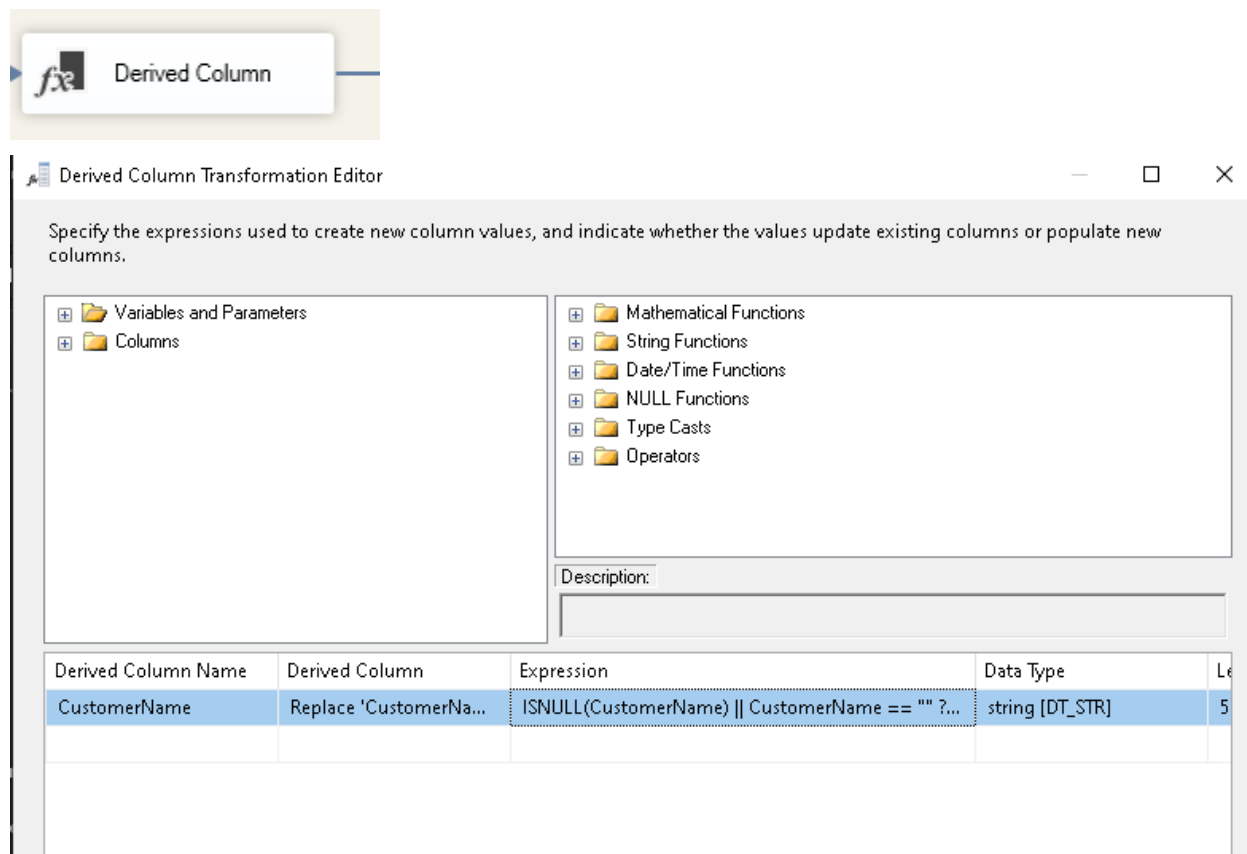
Purpose: Replace NULL names.

Expression:

`ISNULL(CustomerName) || CustomerName == "" ? "Unknown" : CustomerName`

Configuration:

- Derived Column → **Replace CustomerName**



B. Data Conversion (Fix datatype)

Purpose: Convert datatype.

Example:

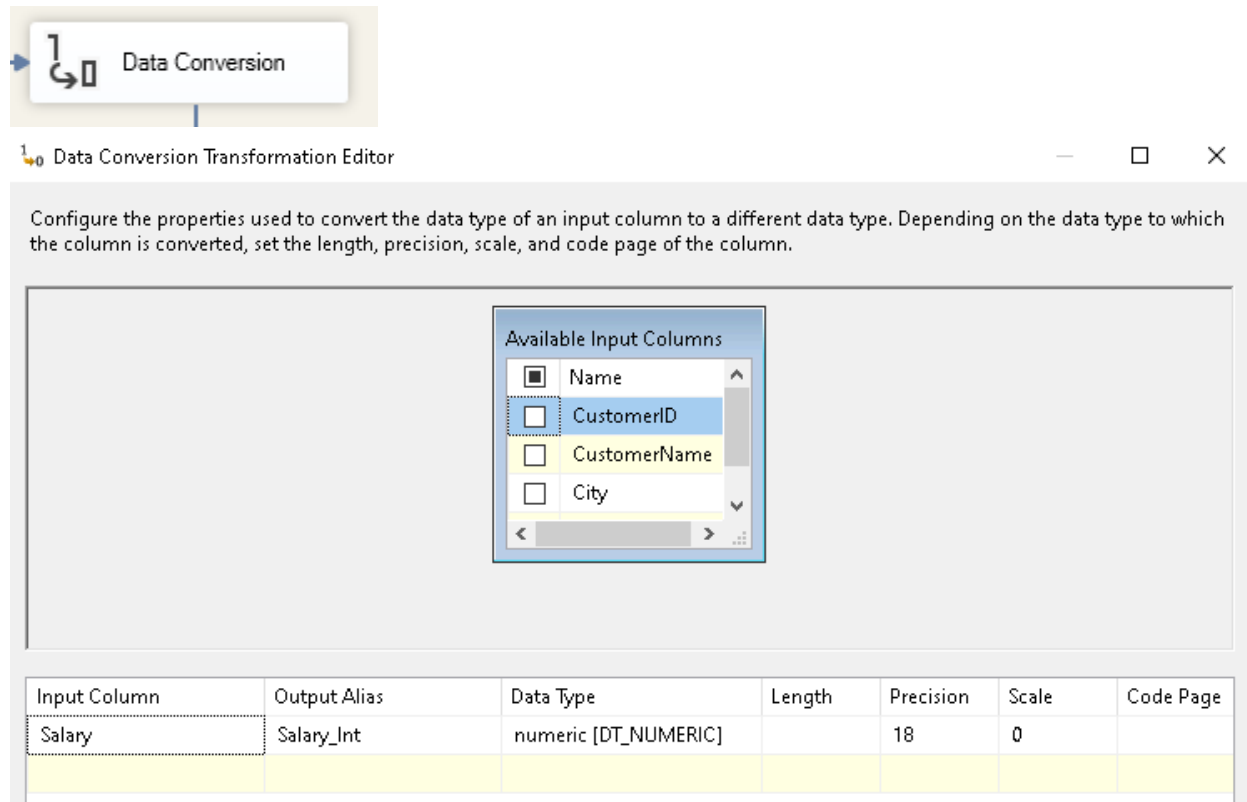
Salary (string) → Salary_int (numeric)

Steps:

1. Open **Data Conversion**
2. Select column **Salary**
3. Choose datatype **numeric (integer)**

Output column example:

Salary_int

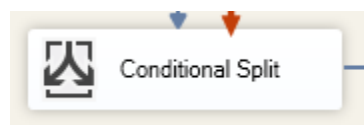


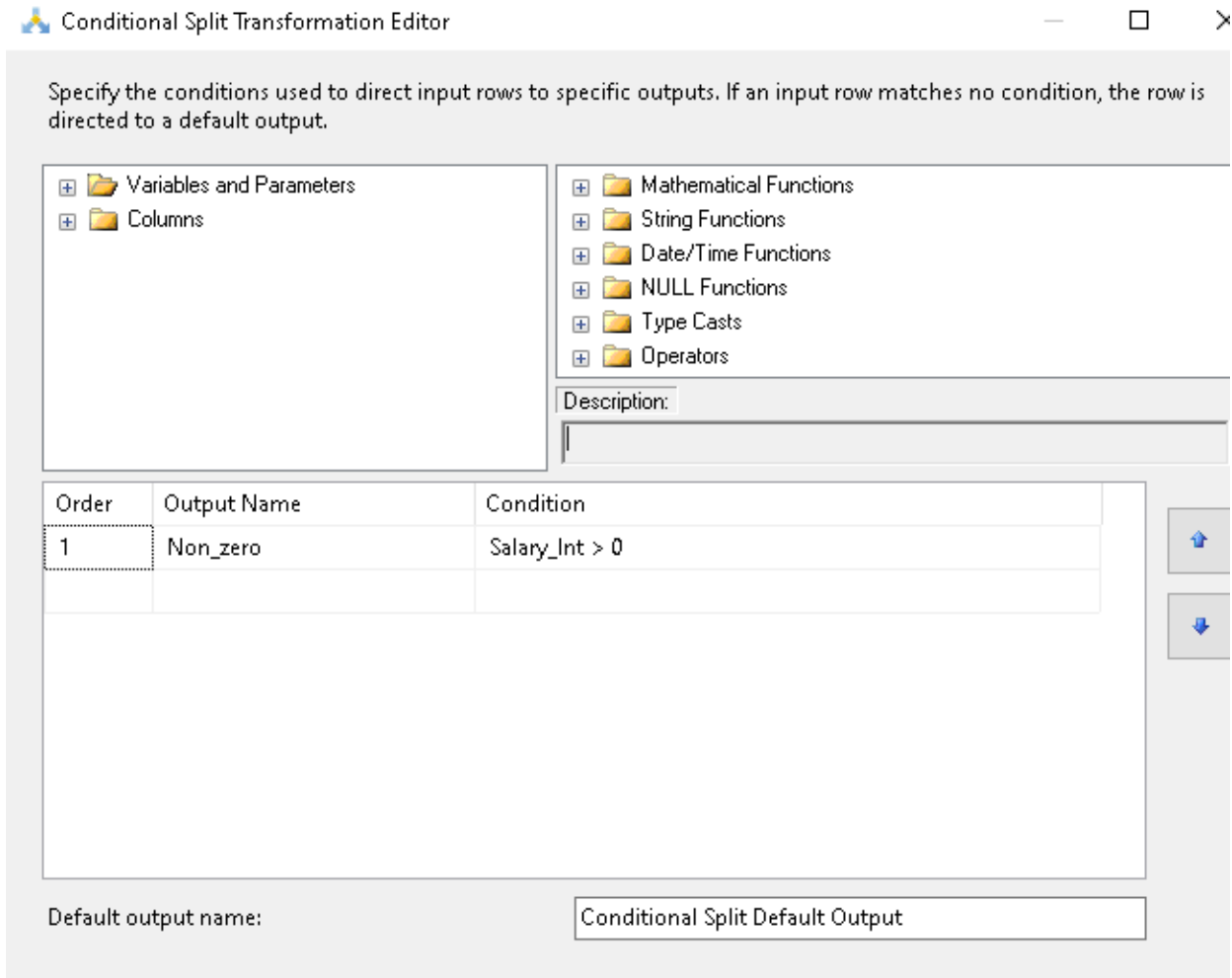
C. Conditional Split (Filter invalid rows)

Purpose: Remove incorrect data.

Example condition:

Salary_int > 0





3 Load to Warehouse (Load)

Add **OLE DB Destination**.

Steps:

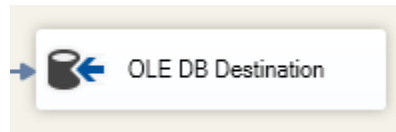
1. Drag **OLE DB Destination**
2. Connect from **Conditional Split** → **ValidRows**
3. Choose database connection
4. Select destination table

Example table:

emp_cleaned

Mapping example:

Source Column	Destination Column
CustomerID	CustomerID
CustomerName	CustomerName
City	City
Salary_int	Salary



OLE DB connection manager:

f726fa6f16e2563.ssis_db.sa New...

Data access mode:

Table or view ▼

Name of the table or the view:

[emp_cleaned] New...

Connection Manager

Mappings

Error Output

Available Input Columns

Name
CustomerID
CustomerName
City
Salary
Salary_Int

Available Destination Columns

Name
CustomerID
CustomerName
City
Salary

Input Column	Destination Column
CustomerID	CustomerID
CustomerName	CustomerName
City	City
Salary_Int	Salary

4 Final Data Flow

Flat File Source



Derived Column



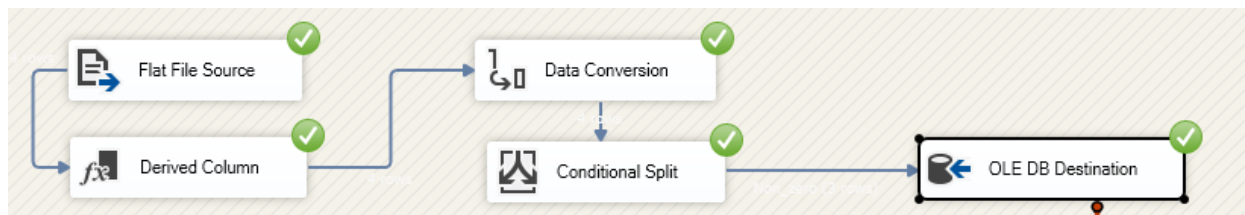
Data Conversion



Conditional Split



OLE DB Destination



Result table:

	CustomerID	CustomerName	City	Salary
1	1	Ravi	Chennai	5000
2	2	Arun	Bangalore	6000
3	3	Unknown	Hyderabad	4500